Journal of Environmental Management 373 (2025) 123424

Contents lists available at ScienceDirect

Journal of Environmental Management

journal homepage: www.elsevier.com/locate/jenvman

Research article

Optimizing the utilization of harvested wood products for maximum greenhouse gas emission reduction in a bioeconomy: A multi-objective optimization approach

Theresa Boiger[a,*], Claudia Mair-Bauernfeind[a], Raphael Asada[a], Tobias Stern[a,b]

[a] Department of Environmental Systems Sciences, University of Graz, Graz, Austria
[b] Wood K Plus—Competence Centre for Wood Composites and Wood Chemistry, Kompetenzzentrum Holz GmbH, Altenberger Straße 69, 4040 Linz, Austria

ARTICLE INFO

Keywords:
Wood utilization
Harvested wood products
Substitution effect
Carbon storage effect
Climate change mitigation

ABSTRACT

Climate change mitigation in a bioeconomy can be attained by increased use of harvested wood products. Thereby, substitution effects can contribute to reducing the Global Warming Potential, and storage effects can prevent direct carbon emissions to the atmosphere. Substitution and storage effects are often only considered as marginal changes. However, an absolute upper limit exists for these effects due to limited harvesting of forests. The maximum emission reduction potential of these two effects depends on the distribution of wood across wood value chains. This study investigates how wood can be distributed optimally across value chains to achieve the maximum climate change mitigation potential by substitution and storage effects, taking a multi-objective optimization approach. Results indicate that sawable wood should be utilized in sawnwood applications to achieve the highest GHG emission reduction, while non-sawable wood contributes to maximal storage effects when used in material applications. The highest substitution effects occur when non-sawable wood is used for energy production. Both substitution and storage effects lead to a substantial mitigation effect, whereby storage effects yield a higher total reduction. The presented optimization method overcomes the limitations of marginal considerations and highlights opportunities for climate change mitigation within these limits. This study emphasizes the importance of taking a systems perspective on climate change mitigation in a bioeconomy and of understanding the limits of these effects regarding environmental policies and management.

1. Introduction

The political vision of bioeconomy encompasses climate change mitigation by relying on renewable resources for industrial production while ensuring their sustainable use and reducing dependence on non-renewables (European Commission, 2018). Besides carbon sequestration in forests, two major effects contribute to climate change mitigation: carbon storage and substitution. Firstly, harvested wood products (HWP) partly store the sequestered carbon during their lifetime and prevent direct carbon emissions to the atmosphere, resulting in what are referred to as storage effects (Geng et al., 2017). Secondly, replacing GHG intensive materials and fossil energies with HWP can reduce GHG emissions, resulting in what is referred to as substitution effects (Leskinen et al., 2018). The combination of these effects is referred to as mitigation effects in this study.

To quantify substitution and carbon storage effects, research previously focused strongly on marginal effects, but not on the maximum mitigation potential. Regarding substitution effects, displacement factors have been used to describe the avoided GHG emissions per unit of wood product replacing a conventional product (Sathre and O'Connor, 2010), but these often do not capture the overall climate change mitigation effect (Myllyviita et al., 2021). Carbon storage effects are calculated as marginal stock changes (net stock addition), and a mitigation effect occurs when the HWP carbon stock size is increased and maintained (Hetemäki et al., 2022). This mitigation effect holds as long as the flow equilibrium is maintained. Many studies have calculated scenarios where a marginal amount of wood is substituted to determine the substitution and carbon storage effects per additional unit of wood used, but have not considered availability of wood (Brunet-Navarro et al., 2021; Dugan et al., 2018; Hurmekoski et al., 2020; Lauri et al., 2021; Rüter et al., 2016).

As the availability of wood is limited, an upper limit exists for

* Corresponding author.
E-mail address: theresa.boiger@uni-graz.at (T. Boiger).

https://doi.org/10.1016/j.jenvman.2024.123424
Received 14 June 2024; Received in revised form 18 September 2024; Accepted 18 November 2024
Available online 28 November 2024

marginal mitigation effects by substitution and carbon storage. Based on sustainable forest management, harvest can only be as high as increment to not reduce the forest area. In Europe, the forest area increased since 1950 (Fuchs et al., 2013) and, e.g., in Austria, harvest only accounted for 89% of the increment between 2016 and 2021 (BFW, 2022). However, increasing the harvest compared to the current level has negative impacts on the overall carbon balance if the reduction in the forest carbon pool is higher than the generated substitution and storage effects (Jasinevičius et al., 2017; Leturcq, 2020; Soimakallio et al., 2021). This aspect becomes especially relevant under climate change conditions, as increasing forest disturbances threaten forest carbon pools (Seidl et al., 2017). Even when the harvest can be increased, there is an upper limit to the availability of wood; therefore, there is a maximum level of substitution and storage effects.

The magnitude of mitigation by substitution and storage effects is determined by how wood is used across wood value chains. The circular economy and cascading utilization concepts indicate that higher mitigation effects can be achieved by utilizing wood in material applications to produce high-value products (Geng et al., 2017; Haberl and Geissler, 2000; Mair and Stern, 2017). However, there is also a mitigation potential for using wood in the energy sector (Poudel et al., 2011; Wolf et al., 2016), and the effects are especially high if fossil fuels are replaced by wood (Koh and Ghazoul, 2008). Substitution effects differ strongly depending on the type of wood product and the industry where substitutes are provided (Boiger et al., 2024a, 2024b; Hurmekoski et al., 2021; Knauf et al., 2015; Sathre and Gustavsson, 2009; Suter et al., 2017). The product lifetime and thus storage effects also differ strongly between industries (Brunet-Navarro et al., 2017). Therefore, information about the distribution of wood across wood-processing industries is required to identify the maximal substitution and storage effects.

To find the maximum mitigation effect, a method is needed to optimize the distribution of wood used across the wood value chains. Several studies have used optimization methods, mainly in the context of optimizing forest management or wood value chains targeting environmental or economic objectives (e.g. Hennigar et al., 2008; Hiltunen et al., 2021; Pukkala, 2011; Raymer et al., 2009; Shabani and Sowlati, 2013; She et al., 2019). As several literature reviews show, most optimization studies on forest or wood value chains examine energy systems (Acuna et al., 2019; Santos et al., 2019; Shabani et al., 2013), but disregard the material usage of wood. The few studies that included material applications of wood in the optimization emphasized recycling or cascade use (Höglmeier et al., 2015; Mehr et al., 2018; Taskhiri et al., 2019). However, none of the studies have examined the optimal distribution of wood for use in material and energy production to maximize the mitigation effect.

The aim of this work is to find the distribution of wood that leads to the maximum mitigation potential in the Austrian wood usage system by substitution and carbon storage effects considering the entire wood value chains. We take a systems perspective on substitution and storage effects and consider system-wide direct and indirect effects stemming from variations in the allocation of roundwood to utilization routes. The wood harvest volumes are kept constant to isolate the effect of changed wood distribution. We determine the maximum reduction potential under the constraint of limited availability by performing an optimization of the wood usage system. We deliberately disregard economic constraints and provide insights into how far we are from realizing this potential given the current economic conditions. Therefore, this paper is intended to answer the following research question: What is the optimal distribution of wood across wood value chains to achieve the maximum climate change mitigation potential by substitution and carbon storage effects?

2. Methods and data

In general, the system of wood usage consists of value chains from the forest to industries. The raw material is processed along the value chain until end products such as paper, energy, or furniture are produced by the respective industries. At this stage, the carbon content of HWPs is accounted for to estimate the storage effect. Within each industry, functionally equivalent products can be made from alternative materials that serve as substitutes for wood. The substitution effect occurs when fossil emissions are avoided by producing a wood-based alternative instead of a (more energy-intensive) substitute (Leskinen et al., 2018).

2.1. Modelling the wood usage system

The WOODSIM model (Boiger et al., 2024a) enables to examine the wood flows in Austria by taking a system dynamics approach (Fig. 1). The value chains from the forest to the industries are modeled in the form of material flows. The harvesting level is considered as constant below the increment level; therefore, the forest carbon pool is not influenced. The same approach was taken by Mehr et al. (2018) and was chosen to isolate the effects of changes in wood distribution on substitution and storage effects and to reduce the complexity of the model. This does not imply that we believe harvest levels will not be subject to changes in the future. Interrelationships between the value chains are included, such as the use of saw residues from sawmills are used in paper production. Within the material flows, quality (non-sawable wood or sawable wood) and wood species (hardwood or softwood) are differentiated. Substitution is included in the model by defining alternative materials for each wood-based product (green box in Fig. 1). For both, wood-based products and substitutes, the Global Warming Potential (GWP) is calculated, and the difference is taken to define the substitution effect. Storage effects were not included in the previous version of the model, as the time span simulated was one year, but were added for the purpose of this study (orange box in Fig. 1).

Several adaptations to the WOODSIM model from (Boiger et al., 2024a) were implemented as described below, which enables the analysis of all parts of the HWP use system, and especially the consideration of the long-term effects. The modeling time span was extended to simulate the wood usage system over a period of 100 years. This is fundamental, as the impact of storage effects becomes evident only when taking a long-term perspective (Dugan et al., 2018; Rüter et al., 2016).

Firstly, the model was adapted to include storage effects. The carbon sequestered in the wood of the HWPs is transferred from the production to the HWP pool. The carbon is stored for the lifetime of the HWP and released again at the end-of-life. The lifetime of HWPs is defined based on the principle of half-lives, meaning that half of the products are still in the product pool after a defined number of years (Brunet-Navarro et al., 2016). The default half-lives from the IPCC guidelines (Rüter et al., 2019) were taken, namely 35 years for sawnwood, 25 years for wood-based panels, and 2 years for paper, paperboard and textiles. After the final use, products go from the HWP pool to energy production.

Secondly, cascading utilization (e.g., paper and wood product recycling) was considered along with the storage effects (see Fig. 1, violet flows). Paper recycling is relevant for substitution and storage effects, as around 50% of the input material for paper production in Austria comes from recycled paper (Austropapier, 2021), and substantial amounts of waste wood are used for panel and energy production (Strimitzer et al., 2020). Regarding the long-term effects, if fewer paper and wood products are produced in one year, their availability for recycling in subsequent years is influenced (Garcia and Hora, 2017). Under the assumption that a constant demand has to be met, additional fresh wood would be needed to compensate for the reduced amount of waste paper or waste wood. Cascade use was previously considered in the model, but not in the form of dynamic flows. Here, the benefit of the system dynamics modeling approach becomes apparent: by taking this approach, feedback loops can be modeled (Homer, 2019; Sterman, 2000), for example, when paper production influences the amount of paper available for recycling, which again influences paper production. In the

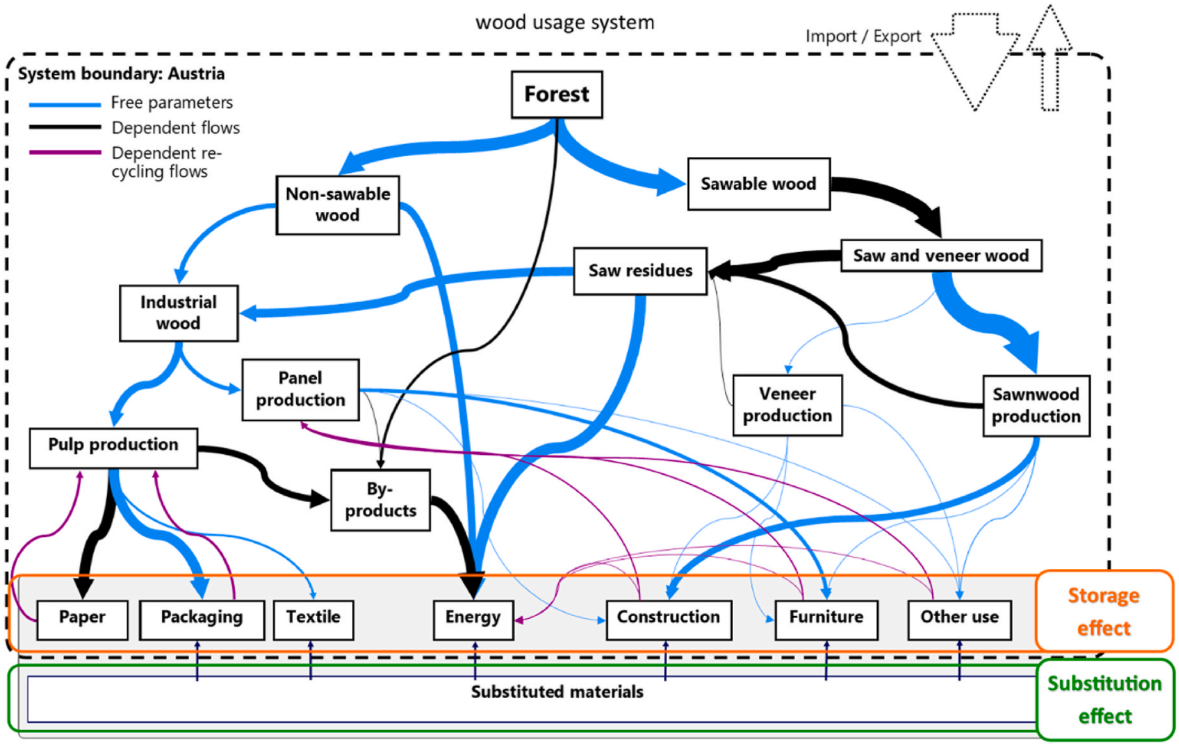

Fig. 1. Wood usage system with blue lines representing free parameters (wood flows that can be varied by the optimization algorithm) and black and violet lines showing dependent flows and dependent recycling flows. (Flows of wood and recycled materials that depend on free parameters are not variable by the optimization algorithm).

model, paper flows from the HWP paper pool back to paper production and to energy production. Waste wood from construction, furniture and other use industries is used for panel and energy production.

Thirdly, the substitution effects were transformed into long-term effects. In the model, substitution effects are calculated for each year of the simulation period. As a simplification, it is assumed that the substitution effect is constant over the simulated period. Details on modeling substitution in WOODSIM can be found in Boiger et al. (2024a). Fourthly, an upper limit for substitution was defined to avoid market saturation. This upper limit is set to avoid a scenario where every product in a particular industry is exclusively made of wood, since this would be neither realistic nor technically possible. As the real maximum potential for wood products in each industry is not known, the following assumption was made: Based on the suggested method for defining the substitution index in the criticality assessment, the maximum additional substitution was defined as 50% of the potential substitutable market share (Blengini et al., 2017). For instance, in the packaging industry, the market share of wood is currently 92% (Statistik Austria, 2018), and the rest (8%) is the potential substitutable market share. Fifty percent thereof, equal to 4% of the market, can be substituted additionally, resulting in a maximum potential market share for wood of 96%. Setting this upper limit allows to increase the market share of wood substantially without saturating the market fully with wood and yields an upper threshold for the optimization (Table 1).

Table 1
Current and maximal market share of wood for upper limit of the substitution effects.

	Current wood market share in %	Assumed maximum wood market share in %
Packaging	92 %	96 %
Textile	8 %	54 %
Construction	7 %	53 %
Furniture	43 %	72 %
Other use	9 %	55 %
Energy	13 %	57 %

The results for the substitution and storage effects are calculated over 100 years and summed up for comparison. To obtain the substitution effects, the results from every year are summed up to give the total substitution effects. To obtain the storage effects, the net addition to the stock for each year over the simulation period is summed up, which gives the additional amount of carbon stored over the whole simulation period.

To assess the maximum mitigation potential, we needed to define which parts of the HWP use system could be adapted to reach an optimal distribution. In the model, the amount of wood used in each industry affects the substitution and storage effects. For this study, the status quo of HWP use was taken as a starting point, and a set of parameters shifting wood between industries determines the two effects. Specifically, depending on the amount of wood utilized per industry, substitutes are provided, thereby reducing GHG emissions (by calculating the difference between additional emissions by wood products and emissions avoided by using substitutes). Furthermore, depending on the amount of wood utilized and the half-lives in each industry, carbon is stored in wood products over a specific time. We disregarded economic constraints in the assessment carried out to identify the maximum mitigation potential. Thus, we assessed a physical rather than an economic potential to demonstrate how much GHG emissions could be reduced under the current conditions and policy landscapes.

The set of parameters that can be varied in the model to perform the optimization consists of all material flows within the HWP system (indicated in blue in Fig. 1). Some material flows such as the saw residues and by-products were disregarded as these depend on the amount of wood used in the respective step. Additionally, the amount of wood used in the production of writing and sanitary paper is fixed; no substitute was assumed here, because the demand needs to be met. With respect to all other material flows in the model, shifts between industries are possible to reduce GHG emissions and increase carbon storage within feasible limits. Detailed information about how this parameter set was implemented is found in section 2.2.1.

2.2. Optimization of the wood usage system

Optimization is a method to define the minimum or maximum of a problem. A problem is usually defined by a goal function, which represents the aspect of the system that should be maximized or minimized (Keller, 2017). If more than one goal is defined, this turns into a multi-objective optimization problem (Dasgupta et al., 1999; Keller, 2019). Free parameters are parameters that can be varied by the algorithm to adapt the goal function and reach a maximum or minimum solution (Kochenderfer and Wheeler, 2019). Additionally, bounds and constraints can be set to define and restrict the problem (Keller, 2017). All options, given a set of free parameters within the bounds and constraints, define the search space which contains the possible solutions from which the algorithm can choose (Keller, 2017). Based on this search space, the algorithm computes the optimal solution.

To solve the optimization problem, an algorithm is needed. Previous studies on optimizing wood or forest value chains mostly used (mixed integer) linear programming (Cambero et al., 2016; Hennigar et al., 2008; Laganovska et al., 2022; Raymer et al., 2009; She et al., 2019; Taskhiri et al., 2019; Zhang et al., 2017) or mixed integer non-linear programming (Shabani and Sowlati, 2013). However, the current problem is complex and characterized by interconnected value chains and feedback loops. Therefore, it cannot be solved linearly. If problems cannot be solved by exact techniques due to their complexity, meta-heuristics are needed which are not problem-specific and can be applied to approximate the optimum (Keller, 2019). Here, deterministic and stochastic optimization techniques exist (Sharma & Kumar, 2022). To solve multi-objective optimization problems, dominance-based algorithms, such as the genetic algorithm, differential evolution, and swarm-based algorithms, are used most often (Q. Liu et al., 2020). One of the most popular multi-objective algorithms is the Non-dominated Sorting Genetic Algorithm II (NSGA II) (Deb et al., 2002), which has been tested and applied widely (Sharma & Kumar, 2022). Multi-objective optimization does not provide a single solution, but a pareto front, which is a set of optimal solutions with regard to all goals (Keller, 2017).

In the context of environmental optimization problems, Life Cycle Assessment and multi-objective optimization strategies can be combined to calculate a range of best solutions and the best alternative for environmental management can then be identified (Azapagic and Clift, 1999; Pieragostini et al., 2012). Höglmeier et al. (2015) and Mehr et al. (2018) combined (Dynamic) Material Flow Analysis with Life Cycle Assessment and optimization to test the suitability of this approach for identifying environmentally optimal solutions for complex wood value chains. In the current study, a System Dynamics Model is used, drawing on Dynamic Material Flow Analysis. The System Dynamics Model is connected to emission flows to assess GHG emissions with respect to substitution effects and the carbon stored with respect to storage effects. The model is combined with multi-objective optimization, which was performed as simulation-based optimization. This approach has also been taken by other researchers, such as Hiltunen et al. (2021), and involves simulating the model and evaluating the result of each run. The two objectives given by the present model are carbon storage and substitution effects. Considering the two effects as separate goals is important to be able to analyze substitution and storage effects separately. Each of the two effects are influenced by different parameters which lead to a distribution of wood over the value chain and therefore to different substitution or storage effects. By defining the problem as multi-objective, a set of solutions that maximize both substitution and storage effects can be found. This gives information on the influence of parameters on each of the two effects. Additionally, the substitution and storage effects might contradict one another and the contradictory parameters can be identified. A single-objective approach would return the optimal results as the sum of substitution and storage effect but not give any information about the separate effects or the influencing parameters. Therefore, combining the two effects might lead to information loss. However, analyzing the set of solutions from the multi-objective optimization still enables to decide which of the solutions is preferable or overall optimal.

2.2.1. Optimization problem

$(1)\ \textit{max substitution}_{\ WOODSIM(x)}$

$(2)\ \textit{max storage}_{\ WOODSIM(x)}$

$(3)\ s.t.\ \ x_i^L \leq x_{i,s} \leq x_i^U\ ;\ x_i^L \leq x_{i,e} \leq x_i^U$

$(4)\ x_{i,s}, x_{i,e} \in \mathbb{R}, i = 1, 2, \ldots, 20\ (\textit{free parameters})$

$(5)\ \textit{inequality constraints}\ (\textit{see Supplementary})$

Equation (1): Multi-objective optimization problem to maximize substitution effects and storage effects.

The multi-objective optimization problem in this study has the following characteristics: Two goals are defined (see Equation (1) (1) and (2)): maximizing substitution effects and maximizing storage effects, both of which are defined based on the WOODSIM model simulation. Each of the goals is defined by the sum of the effects over the simulation period of the model (100 years).

The free parameters (see Equation (1) (3) and (4)) are defined as the shifts between material flows, as explained in section 2.1. The possible shifts in wood usage are summarized in Table 2, where x_i^L , x_i^U are lower and upper bounds for the free parameters to ensure that the results are feasible. The optimization algorithm can vary the amount of wood used in different value chains within these limits. For example, 46% of wood

Table 2
Free parameters (fp) for the multi-objective optimization problem.

Fp (start, end)	Shift from … to …	
$x_{1,s}, x_{1,e}$	forest to non-sawable wood hard	forest to sawable hard
$x_{2,s}, x_{2,e}$	forest to non-sawable wood soft	forest to sawable soft
$x_{3,s}, x_{3,e}$	non-sawable to industrial wood soft	non-sawable to energy soft
$x_{4,s}, x_{4,e}$	non-sawable to industrial wood hard	non-sawable to energy hard
$x_{5,s}, x_{5,e}$	industrial wood to paper production soft	industrial wood to panel production soft
$x_{6,s}, x_{6,e}$	industrial wood to paper production hard	industrial wood to textile production hard
$x_{7,s}, x_{7,e}$	industrial wood to paper production hard	industrial wood to panel production hard
$x_{8,s}, x_{8,e}$	industrial wood to textile production hard	industrial wood to panel production hard
$x_{9,s}, x_{9,e}$	panel production to construction	panel production to furniture
$x_{10,s}, x_{10,e}$	panel production to construction	panel production to other use
$x_{11,s}, x_{11,e}$	panel production to furniture	panel production to other use
$x_{12,s}, x_{12,e}$	sawable wood to veneer production soft	sawable wood to sawnwood production soft
$x_{13,s}, x_{13,e}$	sawable wood to veneer production hard	sawable wood to sawnwood production hard
$x_{14,s}, x_{14,e}$	sawnwood to construction soft	sawnwood to furniture soft
$x_{15,s}, x_{15,e}$	sawnwood to construction soft	sawnwood to other use soft
$x_{16,s}, x_{16,e}$	sawnwood to furniture soft	sawnwood to other use soft
$x_{17,s}, x_{17,e}$	sawnwood to furniture hard	sawnwood to other use hard
$x_{18,s}, x_{18,e}$	veneer to furniture hard	veneer to other use hard
$x_{19,s}, x_{19,e}$	saw residues to industrial wood soft	saw residues to energy soft
$x_{20,s}, x_{20,e}$	saw residues to industrial wood hard	saw residues to energy hard

flows from the stock non-sawable wood are used for industrial purposes and 54% to produce energy (Strimitzer et al., 2020). The limits are set so that a maximum of 46% of the non-sawable wood can be shifted from industrial to energetic wood usage, and a maximum of 54% can be shifted from energetic to industrial wood usage. These limits serve as thresholds, ensuring that no more wood is shifted than is available. The bounds are defined based on the wood flow statistics data (Strimitzer et al., 2020) used as input for the model. Regarding the wood quality, we defined that damaged wood could not be used in sawnwood applications, but sawable wood could be used in all applications. Hardwood is needed for hardwood applications, and softwood is needed for softwood applications.

The free parameters must be variable over time; if these were constant over the simulation period, it would only be possible to identify one optimal distribution of wood flows. As the simulation period is 100 years, the best solution might be a setting where the distribution of wood changes over time. To provide flexibility for the optimization algorithm, therefore, the shifts between material flows could vary over time. To account for these changes in distributions, starting and ending value $(x_{i,s}, x_{i,e})$ for each free parameter were defined, and a linear relation between those two values was established. In this way, the algorithm could identify an increasing, decreasing, or constant amount of wood usage in each application.

Additionally, in cases where wood is distributed to three applications (e.g., sawnwood to construction, furniture and other use), additional restrictions (see Equation (1) (5)) need to be set. The amount of wood shifted cannot exceed the amount of wood available. For example, the amount of wood shifted from furniture and other use to construction use can only be as high as the sum of the amount available for furniture and other use, because this is the maximum amount of sawnwood available. The combination of free parameters, bounds and restrictions defines the search space for the optimization algorithm and, therefore, all possible solutions.

To solve this specific problem in this study, we use the algorithm NSGA II (Deb et al., 2002) which is suitable for multi-objective problems. This metaheuristic has already been tested widely and proved to produce results close to the optimum (Sharma & Kumar, 2022). This is an important consideration with regard to the current problem, as the real optimum is not known. NSGA II is also especially suitable for simulation-based optimization (X. Liu and Bakshi, 2018). To implement NSGA II, the python module pymoo (Blank and Deb, 2020) was used with a population size of 100 individuals, which gives the number of solutions per iteration. The number of generations was 1000, which gives the number of iterations; this was also used as a termination criterion, resulting in 100,000 model evaluations in total.

2.2.2. Clustering of solutions

The solutions were clustered to evaluate and analyze the results. The clustering procedure is used to display patterns in solutions. The pareto front consists of 100 solutions, which are optimal solutions with regard to substitution and storage effects. For each of the solutions, a corresponding set of parameters (free parameters) define the solution. This set of parameters defines the distribution of wood flows. Some of the solutions on the pareto front result in better performance with regard to substitution effects and others, to storage effects. In order to display patterns in these solutions, the sets of parameters were clustered to identify which parameter settings led to which solutions. This revealed how to distribute wood across the wood usage system to achieve the highest storage effects and the highest substitution effects. The *k*-means cluster analysis method was used (Macqueen, 1967). The number of clusters (three) was defined based on the elbow method (Nainggolan et al., 2019).

2.2.3. Sensitivity analysis

A sensitivity analysis of the optimization was performed to demonstrate the sensitivity of the optimal result to varying input parameters. The substitution and storage effect are influenced most by the avoided emissions when replacing substitutes and the half-lives of products, respectively. Therefore, the two parameters were varied. For substitution, the avoided emissions were increased and decreased by 10%, i.e., when 1 m3 of wood is substituted in the model, the emissions avoided are 10% higher or lower. The uncertainty of varying impact data within the WOODSIM model by 10% was already tested in Boiger et al. (2024a). For storage, the half-lives were increased and decreased by 10% for comparability with the sensitivity analysis of substitution. The optimization was performed again for each of the varied parameters to see the effect on the optimal result. The number of generations was reduced to 500 for the sensitivity analysis as the optimization with standard values showed that the improvements of both the result and performance of the optimization algorithm were neglectable after 500 iterations. Results of the sensitivity analysis can be found in the Supplementary Material.

3. Results

3.1. Optimal solutions

The pareto front resulting from the optimization (Fig. 2) describes all solutions that maximize substitution effects and maximize storage effects simultaneously. The status quo is characterized by a substitution effect of 0 and a net stock addition of 0, based on the current distribution of wood use. Starting from this status quo, an improvement in both effects in terms of climate change mitigation could be achieved by applying the optimization procedure. The solution with the maximum substitution effects (sol max substitution) avoided a total of 183 M t CO_2eq emissions over 100 years, while the net stock addition amounts to 111 M t CO_2eq. The solution with the maximum storage effects (sol max storage) depicts avoided emissions of 137 M to CO_2eq with a net stock addition of 330 M t CO_2eq over 100 years. When the substitution effect is highest, the net stock addition that can be achieved is less than the maximum net stock addition possible, and, when the storage effect is highest, the amount of GHG emissions avoided is less than the maximum amount of GHG emissions that could be avoided. All solutions between the solution where maximal substitution effects are seen and the solution where maximal storage effects are seen are considered as optimal with regard to both goals; these illustrate the trade-off between substitution effects and carbon storage effects.

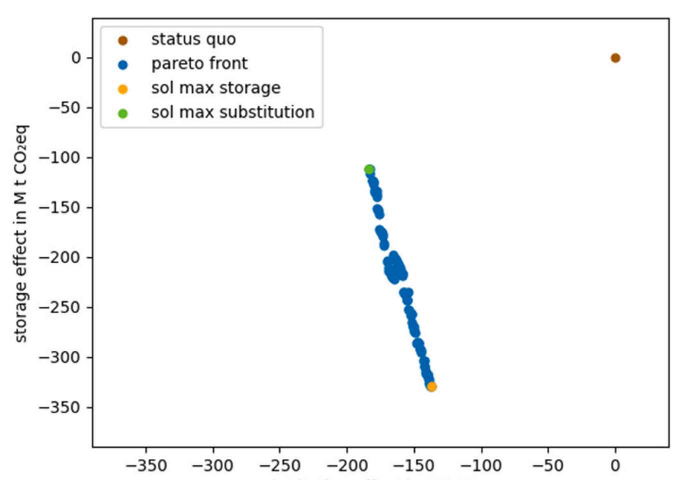

Fig. 2. Pareto front maximizing substitution and storage effects. The solutions highlighted in green and orange depict the solution maximizing storage effects (sol max storage) and the solution maximizing substitution effects (sol max substitution).

The mitigation potential in terms of t CO_2eq is significantly higher for storage effects than for substitution effects (Table 3). Both effects can be combined additively, yielding a total mitigation effect of 467 M t CO_2eq for the solution where maximal storage effects are seen and a total reduction of 295 M t CO_2eq for the solution where maximal substitution effects are seen over the 100-year period as compared to the status quo. Regarding the intermediate solutions, the total mitigation effect lies between the two solutions with regard to all trade-off solutions, which indicates that the solution with the highest storage effects has the highest mitigation effect.

3.2. Parameter settings for optimal solutions

For each solution on the pareto front, a set of parameters (values for free parameters) exists, that leads to the respective solution. To analyze the pareto front, the sets of parameters were clustered into three groups. The set of parameters for all solutions is displayed in Fig. 3b, and the corresponding solutions on the pareto front is shown in Fig. 3a. The set of parameters is normalized between the bounds for each parameter, where 0 is the lower bound and 1 is the upper bound. Cluster 1 contains parameter settings that maximize the storage effect, while cluster 2 contains parameter settings that maximize the substitution effect. Cluster 0 contains all intermediate settings. Some parameter settings display a clear direction towards the lower or upper bound for solutions in all clusters. For example, the values for $x_{1,s}$ and $x_{1,e}$ (i.e., the starting and ending values for free parameter 1) appear at the lower bound in all clusters, indicating a general preference for this shift direction no matter whether substitution effects or storage effects are maximized. When examining other parameters, we see that parameter settings close to the upper or lower bound yield different results. For example, if we examine $x_{3,s}$ and $x_{3,e}$ (i.e., the starting and ending values for free parameter 3), we see that the values for cluster 1 are close to the upper bound, providing a solution with a maximal storage effect, and the values of cluster 2 are close to the lower bound, providing a solution with maximal substitution effect.

The parameter settings shown in Fig. 3b indicate how wood should be shifted and used to optimize the substitution and storage effects. Table 4 shows the free parameters and their trends towards the upper or lower bounds (Fig. 3b), as well as how these influence the two effects. According to the optimization, some clear trends (shaded in grey) are visible. Roundwood (from the forest, x_1, x_2) should be shifted to sawable wood where possible. Industrial wood should preferably be used for panel production (x_5, x_7, x_8) and rather for textile production than paper production (x_6). Sawable wood should be used for sawnwood production instead of veneer production where possible (x_{12}, x_{13}). If we examine other parameters, we see that shifting toward the upper bound tends to increase the storage effect (shaded in violet) and shifting toward the lower bound tends to increase the substitution (shaded in turquoise). Specifically, using non-sawable wood in industrial wood applications increases the storage effect, while using it for energy production increases the substitution effect (x_3, x_4). Using panels in construction increases the storage effect, but using them in the other use industry causes a substitution effect. Using panels for furniture production results in both storage and substitution effects (x_9, x_{10}, x_{11}). The same applies to soft sawnwood (x_{14}, x_{15}, x_{16}). Mixed effects are seen for hard sawnwood and veneer (x_{17}, x_{18}). Like non-sawable wood, using saw residues in industrial wood applications increases the storage effect, while using them in energy production increases substitution effects (x_{19}, x_{20}).

Table 3
Substitution and storage effects for solutions with maximal storage and maximal substitution effects.

	Storage effect (M t CO_2eq)	Substitution effect (M t CO_2eq)	Total reduction (M t CO_2eq)
Status quo	0	0	0
Sol max storage effect	−329.51	−137.10	−466.61
Sol max substitution effect	−111.45	−183.27	−294.72

3.3. Maximal substitution and storage effect

Each set of parameters shown in Fig. 3b reflects a corresponding distribution of wood across the wood usage system. The free parameters indicate where wood is being shifted from one industry to another. If wood is shifted to a wood flow, the amount of available wood increases there and decreases in another flow. Each optimal solution is associated with a set of free parameters and a wood distribution. Below, we analyze the two solutions with the highest storage (sol max storage) and substitution effects (sol max substitution), using these as examples to demonstrate the distribution of wood for each solution. All other optimal solutions lie between those two points and result from intermediate distributions of wood.

The changes in wood flows over the system for both solutions with maximum storage and substitution effects are depicted in Fig. 4. The amount of wood for sawnwood is increased in both solutions which also increases the amount of saw residues. In the solution with maximum storage, saw residues as well as non-sawable wood are increasingly used for panel production which is then mostly distributed to construction. In the solution with maximum substitution, the saw residues are partly used for panel production and partly for energy where also most of the non-sawable wood is used. In both cases the use of wood for veneer and pulp production is reduced.

Fig. 5 shows the resulting difference in wood use per industry as compared to the status quo over time. In the solution maximizing storage effects, more wood is mainly used in construction and other use industries, while less wood is used for packaging and energy compared to the status quo. In construction, the increase is due to the additional use of panels, while the amount of softwood used is even reduced. Instead, the soft sawnwood is employed in applications assigned to the other use and furniture categories, and the number of panels used in these industries also is higher. The amounts for packaging and textiles are reduced. Additionally, wood is diverted from energy production which results in a substantially lower volume of wood being used.

In the solution maximizing the substitution effects, the wood usage in furniture, energy and other use industries is increased, while wood usage for construction, textiles and packaging is decreased. In other uses, the wood usage is increased mainly due to the use of panels. Again, the use of panels in construction and furniture is higher, while the use of soft sawnwood for construction is lower. Even less wood is used for packaging and textiles. The amount of wood used for energy is higher than in the status quo due to an increasing use of non-sawable wood.

These wood distributions lead to the following substitution and storage effects (Fig. 6). In the solution maximizing storge effects, substitution effects occur due to increased wood usage in construction, furniture and other use. At the same time, GHG emissions in the packaging, textile and energy industries increase, but still lead to an overall reduction of 137 M t CO_2eq over 100 years as a result of substitution. The construction and other use pools increase substantially in this solution, leading to a total storage effect of 330 M t CO_2 eq. In the solution maximizing substitution effects, the GHG emissions decrease due to higher wood usage in other use and furniture industries and increase in packaging, textile and construction, leading to a total GHG emission reduction of 183 M t CO_2 eq. The storage effect seen here is mainly caused by an increase in the other use and furniture pools. Although the construction pool is smaller, the total change in storage effect is 111 M t CO_2eq.

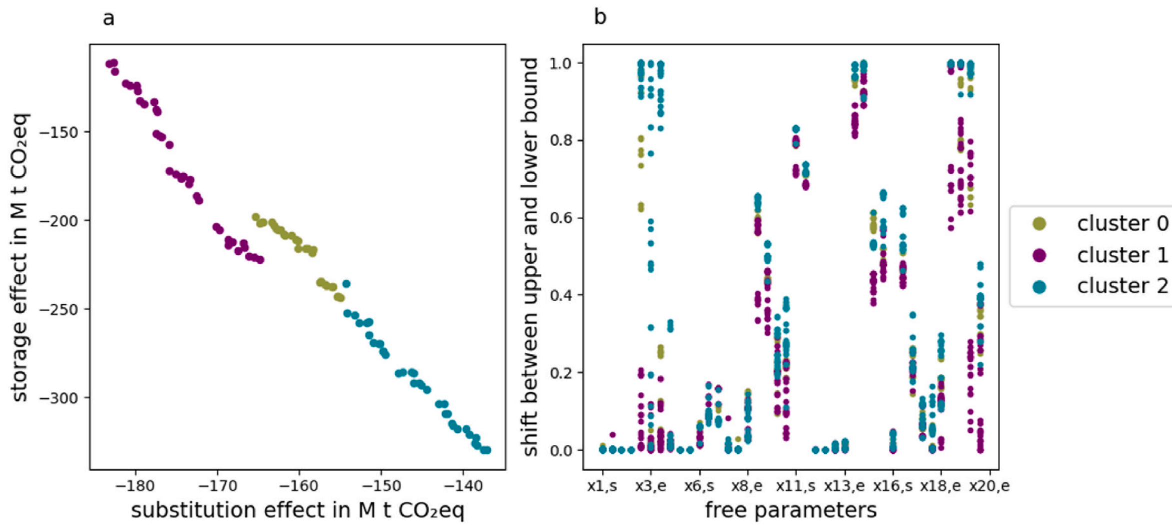

Fig. 3. Clustered pareto front and corresponding setting of free parameters. 3b shows the starting (s) and ending (e) values for each free parameter from 1 to 20 (with omissions in the x-axis labels due to space constraints).

Table 4
Trend in free parameters. Clear shift trends to achieve optimality are shaded in grey. Shifts trends increasing storage effects are shaded in violet, and shift trends increasing substitution effects are shaded in turquoise.

Fp (start, end)	Shift from … to …	
	to (+) upper bound	to (-) lower bound
$x_{1,s}, x_{1,e}$	forest to non-sawable wood hard	forest to sawable hard
$x_{2,s}, x_{2,e}$	forest to non-sawable wood soft	forest to sawable soft
$x_{3,s}, x_{3,e}$	non-sawable to industrial wood soft	non-sawable to energy soft
$x_{4,s}, x_{4,e}$	non-sawable to industrial wood hard	non-sawable to energy hard
$x_{5,s}, x_{5,e}$	industrial wood to paper production soft	industrial wood to panel production soft
$x_{6,s}, x_{6,e}$	industrial wood to paper production hard	industrial wood to textile production hard
$x_{7,s}, x_{7,e}$	industrial wood to paper production hard	industrial wood to panel production hard
$x_{8,s}, x_{8,e}$	industrial wood to textile production hard	industrial wood to panel production hard
$x_{9,s}, x_{9,e}$	panel production to construction	panel production to furniture
$x_{10,s}, x_{10,e}$	panel production to construction	panel production to other use
$x_{11,s}, x_{11,e}$	panel production to furniture	panel production to other use
$x_{12,s}, x_{12,e}$	sawable wood to veneer production soft	sawable wood to sawnwood production soft
$x_{13,s}, x_{13,e}$	sawable wood to veneer production hard	sawable wood to sawnwood production hard
$x_{14,s}, x_{14,e}$	sawnwood to construction soft	sawnwood to furniture soft
$x_{15,s}, x_{15,e}$	sawnwood to construction soft	sawnwood to other use soft
$x_{16,s}, x_{16,e}$	sawnwood to furniture soft	sawnwood to other use soft
$x_{17,s}, x_{17,e}$	sawnwood to furniture hard	sawnwood to other use hard
$x_{18,s}, x_{18,e}$	veneer to furniture hard	veneer to other use hard
$x_{19,s}, x_{19,e}$	saw residues to industrial wood soft	saw residues to energy soft
$x_{20,s}, x_{20,e}$	saw residues to industrial wood hard	saw residues to energy hard

4. Discussion

4.1. Optimal distribution of wood

The optimal distribution of wood across the system under limited availability strongly depends on the displacement factors and half-lives in each industry, as these influence the magnitude of the substitution and storage effects. In general, to maximize the two effects simultaneously, wood should be used for industries with high displacement factors and long product lifetimes (half-lives). The displacement factors

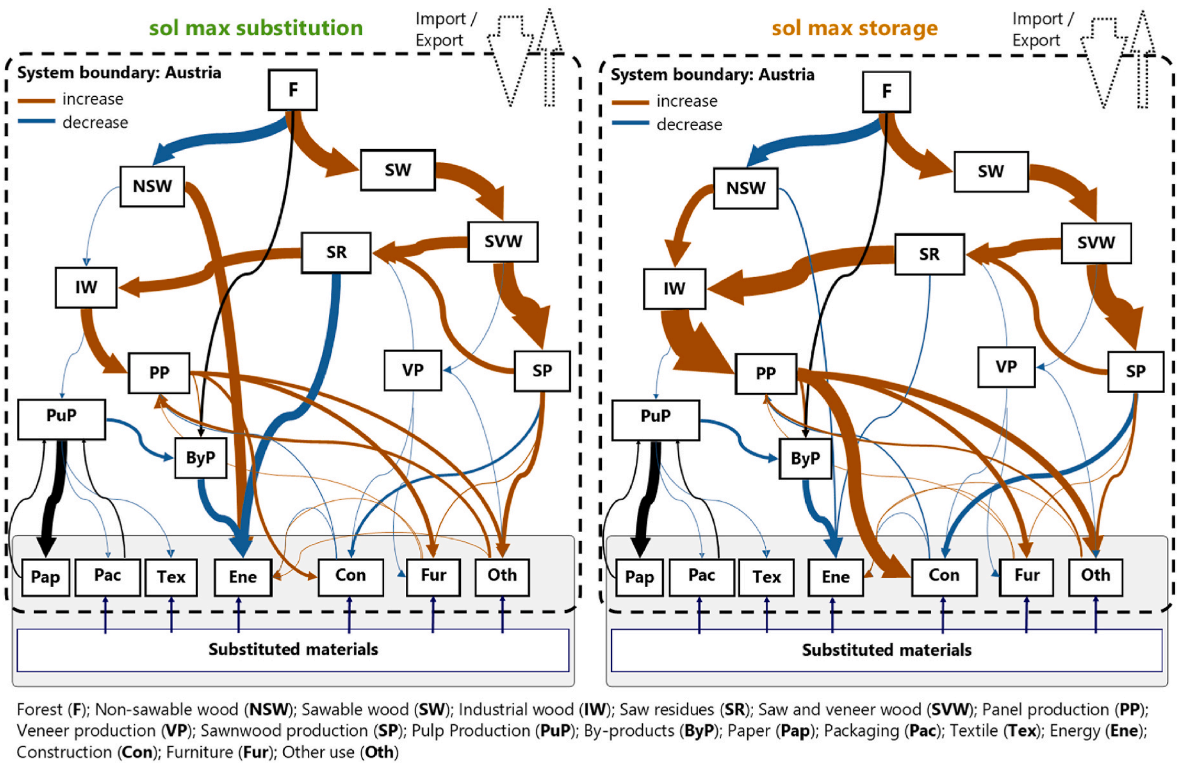

Fig. 4. Optimized wood flows for maximum substitution and storage effects.

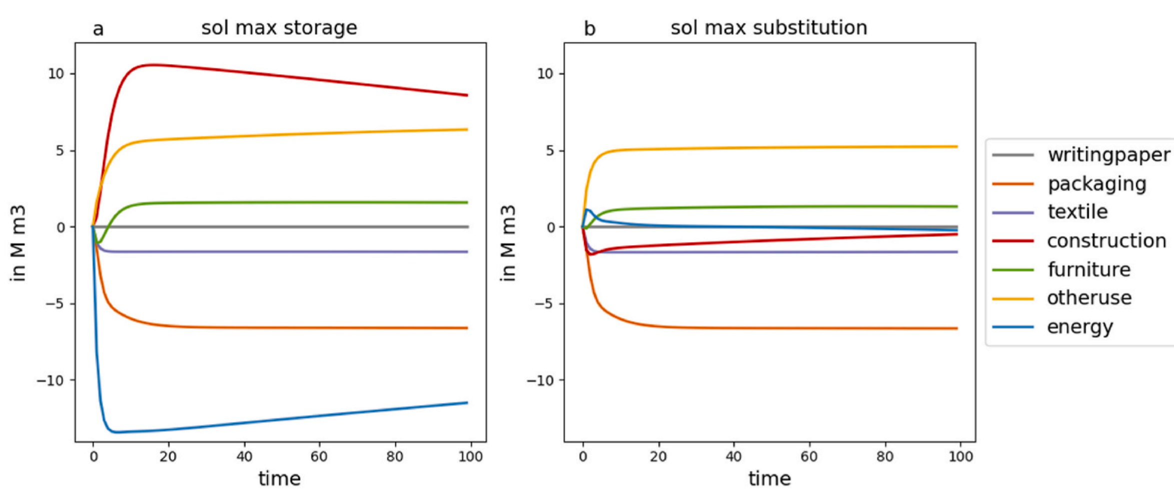

Fig. 5. Difference in amounts of wood per industry for the solution with maximum storage effect (sol max storage) and the solution with maximum substitution effect (sol max substitution) as compared to the status quo.

and half-lives for this study are displayed in Table 5. The shown displacement factors are a result of the calculations and not an input for the substitution. In the optimal solutions, sawable wood is used in sawnwood applications wherever possible. This is in line with literature; Brunet-Navarro et al. (2021) and Hurmekoski et al. (2020) demonstrated that mitigation effects are higher when wood is used for high quality applications. Consequently, sawnwood is mostly used in the furniture industry and in applications assigned to the category of other uses. It seems unintuitively that sawnwood is not preferably used in construction in the optimal solution because many studies have demonstrated a substantial mitigation potential for sawnwood in the construction industry (Cordier et al., 2021; Kalt, 2018; Myllyviita et al., 2022; Schulte et al., 2023; Werner et al., 2005; Zeug et al., 2022). Although the longest half-life would indeed be in the construction sector, the furniture and other uses industries exhibit a higher displacement factor. Additionally, on the system level, the shorter half-life of furniture and other use products results in the production of more waste wood becoming available in the same time period. Waste wood is an important input for panel and energy production, which, in turn, contributes to substitution effects and increases the carbon storage of panels. Therefore, sawable wood is preferably used for furniture and other uses under the conditions assumed in this study.

The same applies to non-sawable wood, which should preferably be used in industries with high displacement factors and long product lifetimes. The packaging and textiles industries have lower displacement factors and half-lives compared to other industries, which explains why the amount of wood is reduced strongly here in the optimal solutions. Large amounts of non-sawable wood are used for either panel

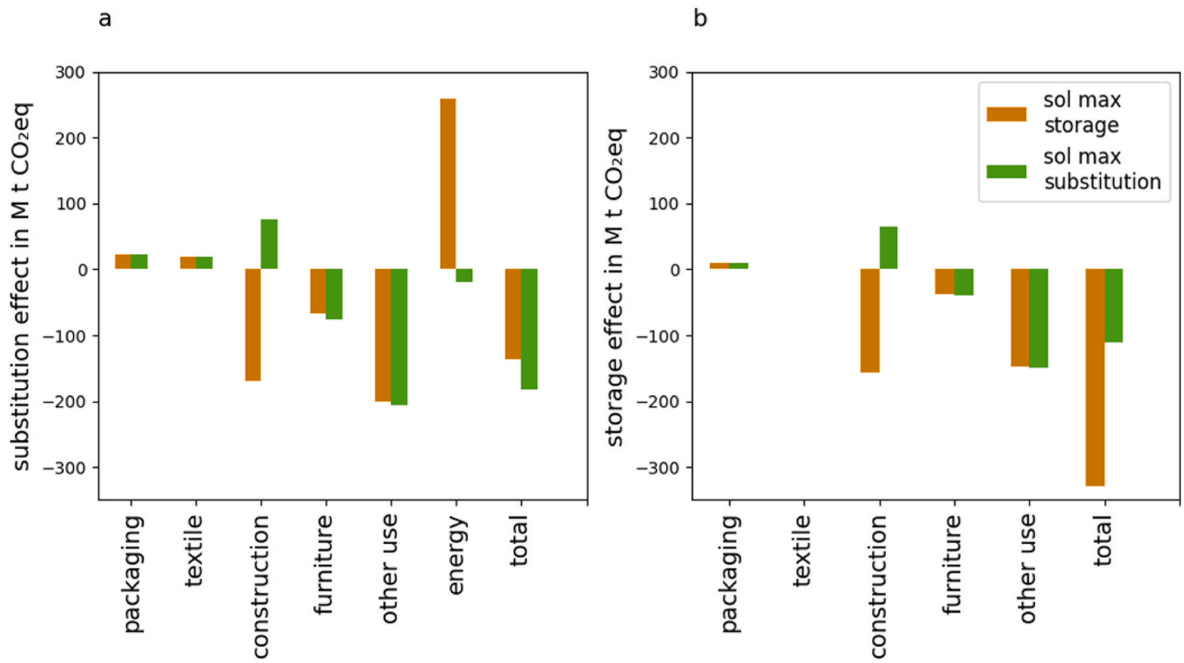

Fig. 6. Difference in GWP in terms of the substitution (left) and storage effects (right) between the status quo and the solution with maximum storage (sol max storage) and the solution with maximum substitution (sol max substitution), respectively.

Table 5
Displacement factors and half-lives in this study.

Half-lives and displacement factors used in the model		
Industry	Half-life in years (Rüter et al., 2019)	Displacement factor tC/tC (Boiger et al., 2024a)
Packaging	2	0.0680
Textile	2	0.1766
Construction	35	0.5421
Furniture	25	0.5907
Other uses	25	0.6931
Energy	0	0.3713

production or energy production. On the one hand, using more non-sawable wood for panel production implies that more panels are used in the construction, furniture and other use industries, where the product half-lives and the displacement factors are the highest as compared to other industries. Especially in construction, high volumes of panels compensate for the diverted sawnwood, leading to high storage effects due to the long product lifetimes (Dugan et al., 2018). In this case, the substitution effect is lower, because the wood unavailable for energy production has to be compensated by (fossil) substitutes. These GHG emissions counteract the substitution effect seen in the construction sector. On the other hand, using non-sawable wood for energy production suggests a lower usage of (fossil) substitutes and a higher substitution effect, even when considering the counteracting effect in the construction sector. In this case, the storage effect is significantly lower due to the absence of carbon storage in energy production. Therefore, using non-sawable wood to produce panels for construction contributes mainly to the storage effect, while using it for energy production contributes mainly to the substitution effect.

When examining the optimal distribution of wood across the wood usage system, we considered that the distribution always results from a combination of parameters. Changing single parameters and therefore shifting wood from just one industry to another might draw an incomplete picture. For instance, shifting wood to panel production for use in the construction industry does not automatically reduce GHG emissions; instead, this shift acts in combination with the use of sawnwood in furniture and other use industries. Potentially, a mitigation effect can only be achieved by combining several parameters.

Some assumptions potentially influenced the optimization result found in this study regarding both the substitution and storage effects. On the one hand, in terms of wood product lifetimes, the default half-lives for sawnwood, wood-based panels and paper suggested by Rüter et al. (2019) were used in this study. In reality, these half-lives will differ strongly, depending on the specific wood product (Brunet-Navarro et al., 2017) and are also likely to vary over time (Braun et al., 2016). Wang and Haller (2024) found that extending the product lifespans improved wood efficiency, and Mehr et al. (2018) showed that the lifetimes in material applications of wood strongly influenced the overall environmental performance. Therefore, the maximal storage effect found might be even higher when half-lives are extended or time-varying.

On the other hand, the displacement factors in this study were assumed to be constant over the simulation period. The displacement factors are derived from the substitution within the model, and detailed information is provided in Boiger et al. (2024a). Here, we assumed that the impact in terms of GHG emissions from both wood products and substituted products did not change over time, resulting in constant displacement factors. However, the impacts of different products are likely to change in future due to, for example, the carbon neutrality goals set by the European Union (European Union, 2020). Additionally, market dynamics might influence the hypothetical substitution assumed in displacement factors (Hurmekoski et al., 2021). Mehr et al. (2018) demonstrated that the environmental performance of the system is affected by the substitution benefits associated with different material recycling options. Brunet-Navarro et al. (2021) showed clearly that the potential substitution effect of wood will decrease in future when applying dynamic substitution factors. As a consequence, the substitution effects might be even lower when dynamic substitution factors are used. It is up to further research to assess the impact of dynamic displacement factors on the results.

4.2. Substitution vs. storage effects

When optimizing for climate change mitigation, maximizing substitution effects and storage effects can be contradictory goals. The maximum mitigation potential from the substitution effects seen in this study is 183 M t CO_2eq, while the storage effect potential is 330 M t CO_2eq. These results indicate that storage effects have a higher reduction potential. However, is it not possible to achieve the maximum reduction potential with respect to both goals simultaneously. There are trade-offs between the two goals which are shown by the pareto front of the solution. Previous studies described trade-offs between carbon sequestration in forests and harvesting wood for substitution effects and carbon storage in HWPs (Grassi et al., 2021; Leturcq, 2020; Seidl et al., 2007; Soimakallio et al., 2021; Taeroe et al., 2017). Little attention, however, has been directed toward the contradictory of substitution and storage effects seen after harvesting. Depending on the distribution of wood within the wood usage system, either substitution or storage effects can be maximized. When aiming for the maximal mitigation effect, the solution with optimal storage effects is preferable, and a reduction of 467 M t CO_2eq can be achieved.

Increasing storage effects and decreasing substitution effects in the future could further highlight the significance of storage effects. When product lifetimes or cascade use are increased as targeted in the bioeconomy strategy (European Commission Directorate-General for Research and Innovation, 2012; Federal Ministry Republic of Austria, 2019), the storage effects will increase. In contrast, when product GHG emissions are decreased – one of the European Union's emission reduction goals (European Union, 2020) – the substitution effects will tend to shrink. Still, these two effects cannot be seen independently, as they are both influenced by the amounts of wood used in specific industries; thus, interrelations and trade-offs have to be considered. Mehr et al. (2018) indicated that longer lifetimes, and the resulting storage benefits may delay wood incineration processes in the future, meaning that the substitution benefits in energy production might be reduced or even eliminated. Similarly, as shown in this study, the longer lifetime of products in the construction sector is not necessarily more beneficial than the shorter lifetimes of furniture and products for other uses, since increasing the long-term uses of wood temporarily reduces the supply of waste wood and delays the associated substitution effects. Previous studies have already shown that optimizing cascade use can generate a substantial mitigation effect (Höglmeier et al., 2015; Mehr et al., 2018). The sensitivity analysis demonstrates that a variation of input parameters like avoided emissions and half-lives affects the magnitude of the maximum reduction potential, but does not strongly affect the parameter settings and therefore the distribution of wood over the system for optimal solutions. However, further sensitivity or scenario analysis of the optimization needs to be conducted to make statements about the optimum wood distributions under different future assumptions, taking into account changes in energy mixes and cascade use.

4.3. Limitations

Several limitations of the model have to be considered: Uncertainties exist regarding the data input for the model, e.g., potential measurement errors. These were addressed in a previous study by carrying out Monte Carlo simulations, which uncovered some uncertainties associated with substitution effects – especially in the energy sector – but fewer uncertainties in the material-based industries (Boiger et al., 2024a). Both substitution and storage effects analyzed in this study, however, are difficult to study empirically. Substitution processes are notoriously hard to observe, but storage effects are physically observable; nevertheless, data on HWP pools are based on extrapolations from historical data and hence uncertain. Most Life Cycle Assessment studies and available data place a focus on the production phase, but both substitution and storage phenomena refer to use phase impacts for which data availability is low and uncertainty is high.

While the geographical scope of this study was Austria, the optimal distribution of wood and the maximal GHG emission reduction potential could vary in other countries due to differences in structural conditions in the wood-processing industries, such as different wood flow distributions or a lack of certain industries (e.g., textile production). Changes in the forest carbon pool were not modeled, as the focus of this study was on the systemic wood usage aspects. The harvest was kept constant when performing the optimization, thereby avoiding influences on the forest carbon pool. The same approach was taken by Mehr et al. (2018), who kept the harvest constant to optimize a cascade use system. The maximum substitution potential within each industry was defined as 50% of the potential substitutable market share (Blengini et al., 2017), as the real maximum potential was unknown. The optimal solution might look different if other maximal market shares were used. Nevertheless, the wood distribution trends will be similar, while the magnitude of the storage and substitution effects may differ.

The model has been developed to identify a maximum GHG emission reduction potential and the resulting optimal wood distribution across various uses, irrespective of economic factors such as market prices or constrained demand. Introducing economic considerations would narrow the search space, resulting in sub-optimal solutions, particularly in the absence of financial incentives to support climate-friendly decisions, such as effective carbon pricing. While disregarding economic constraints was necessary to achieve the aim of the study, taking this approach prevented us from making any statement about the extent to which the identified potential can actually be realized.

Some limitations affected the optimization procedure as well. The optimization algorithm used (NSGA II) is heuristic in nature; therefore, it provides an approximation and does not guarantee to yield the real optimal solution. However, NSGA II has been tested and used widely to obtain results close to the optimum solutions for standard problems (Sharma & Kumar, 2022). The simulation period for the model was defined as 100 years. Optimizing the model with a shorter or longer simulation period might change the optimal result. However, our results suggest that the transition period from the current distribution of wood to the optimal one is only about 20 years. Thereafter, the distribution remains constant, indicating that the optimal result would not change significantly if a different simulation period were used.

Regarding the free parameters for the optimization, all material flows in the wood usage system (except dependent flows) are adaptable. The recycling flows were dependent and therefore, no improvement in recycling efficiency or cascade use was possible. The aim in this study was to investigate the distribution of the available wood rather than to improve recycling or cascade use. However, Höglmeier et al. (2015) and Mehr et al. (2018) both found that optimizing cascade use reduced GHG emissions in their studies. Examining additional substitution and storage effects due to increased recycling or cascade use is thus a topic for further research.

4.4. A system perspective on bioeconomy

The maximum GHG reduction potential shown in this study supports the hypothesis that marginal substitution and storage effects are limited if a system perspective is applied. The maximum potential substitution effects are relatively small compared to the maximal mitigation potential and might decrease in the future. Still, replacing non-renewable materials and energies with wood is preferable (Federal Ministry Republic of Austria, 2019). In contrast, the potential storage effects occurring in a bioeconomy can substantially influence climate change mitigation efforts. The average maximal mitigation effect per year is 4.67 M t CO_2eq. Compared to Austria's total annual GHG emissions (77.5 M t CO_2eq in 2021 (Anderl et al., 2023)), the maximal mitigation effect corresponds to around 6% of Austria's emissions. Our study provides strong support for the argument that implementing circular economy concepts and promoting improved recycling and cascade use are necessary to further reduce the overall GHG emissions of the wood usage system (Höglmeier et al., 2015; Mehr et al., 2018). However, such concepts can only be applied effectively if the corresponding framework conditions are in place. For instance, carbon leakage must be prevented, where the emissions avoided by carbon storage and substitution are released by other sectors (Harmon, 2019). Similarly, accompanying measures are required to ensure that climate change mitigation efforts are not nullified by uncontrolled consumption increases (Khmara and Kronenberg, 2020).

5. Conclusion

Substitution and storage effects are often considered as marginal effects, neglecting that an upper limit to climate change mitigation exists due to the limited availability of wood. This study assesses the maximum mitigation potential of wood by optimizing its distribution across various value chains. The multi-objective optimization applied in this study demonstrates how wood should be distributed to maximize both the substitution and the storage effects over a period of 100 years:

- Compared to the status quo, both substitution and storage effects can be reduced. However, the maximal substitution effect and maximal storage effect cannot be achieved simultaneously, because they require different wood distributions across industries. This demonstrates that upper limits to marginal effects exist.
- The climate change mitigation potential of storage effects is higher than that of substitution effects. While substitution effects are outweighed by effects and counter effects (e.g., a simultaneous GHG emission reduction in construction, furniture and other use and a GHG emission increase in energy production), storage effects have the potential for substantial mitigation when wood is used in long-lived products.
- Wood is used optimally when it is used in high-quality applications whenever possible. When it is used for energy production, substitution effects are maximized, and when it is used in material applications, storage effects are maximized.

As the reduction potential of substitution effects is lower than that of storage effects, substitution rather than storage effects seem to support the optimization of the current wood usage system. Substitution seems to contradict the prevailing economic system less strongly (e.g., green growth, weak sustainability) and is easier to implement. While substitution is supported by, e.g., carbon taxation, increasing the storage effect would require other policy measures. For instance, extending average product service lives can be considered as a sufficiency pathway that contradicts current economic growth paradigms. The calculations presented here, however, are based on the current wood usage system and the difference between the reduction potential of storage and substitution effects; therefore, the results may also just reflect the physical potential of the two.

Future research needs to assess the maximum reduction potential of the two effects under changing future conditions, such as a transition towards renewable energy-based production, prolonged service lives, or fostered cascade use. The social consequences of changes in wood usage should be considered and could be examined as a separate goal in the optimization process. Furthermore, the potential effects of policy measures and innovations, particularly those that support storage effects, should be investigated to take actions towards converging to maximal climate change mitigation.

CRediT authorship contribution statement

Theresa Boiger: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Claudia Mair-Bauernfeind:** Writing – review & editing, Visualization, Validation, Conceptualization, Writing – original draft. **Raphael Asada:** Writing – review & editing, Visualization, Validation, Conceptualization, Writing – original draft. **Tobias Stern:** Writing – review & editing, Validation, Supervision, Conceptualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The project CARpenTiER “Modelling, Production and further Processing of Eco-Hybrid Structures and Materials” is funded within the framework of COMET - Competence Centers for Excellent Technologies by BMK, BMAW and the Province of Styria. The COMET program is managed by the Austrian Research Promotion Agency (FFG, 882509) and the Styrian Business Promotion Agency (SFG, 1,000,065,896). The authors are grateful for this research funding as well as for the co-financing by the companies ACstyria Mobilitätscluster GmbH, BASF SE, BASF Polyurethanes GmbH, DYNAmore Gesellschaft für FEM Ingenieurdienstleistungen mbH, evon GmbH, Fill Gesellschaft m.b.H, Forst-Holz-Papier, Glanzstoff Management Consulting GmbH, Holzcluster Steiermark GmbH, IB STEINER, Klumpp Coatings GmbH, LEAN Management Consulting GmbH, Pfeifer Holding GmbH, Volkswagen AG und Weitzer Woodsolutions GmbH.

Funded by the Austria Research Promotion Agency under the COMET Programme No. 892416.

The authors acknowledge the financial support by the University of Graz.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2024.123424.

Data availability

The data used for this study are described within the Methods Section and the Supplementary Material.

References

Acuna, M., Sessions, J., Zamora, R., Boston, K., Brown, M., Ghaffariyan, M.R., 2019. Methods to manage and optimize forest biomass supply chains: a review. Current Forestry Reports 5 (3), 124–141. https://doi.org/10.1007/s40725-019-00093-4.

Anderl, M., Colson, J., Gangl, M., Kuschel, V., Makoschitz, L., Matthews, B., Mayer, M., Mayer, S., Moldaschl, E., Pazdernik, K., Poupa, S., Purzner, M., Rockenschaub, A.K., Roll, M., Schieder, W., Schmidt, G., Schodl, B., Schwaiger, E., Schwarzl, B., et al., 2023. Austria's National Inventory Report 2023 - Submission under the United Nations Framework Convention on Climate Change.

Austropapier, 2021. Austropapier Branchenbericht 2020/21.

Azapagic, A., Clift, R., 1999. Life cycle assessment and multiobjective optimisation. J. Clean. Prod. 7 (2), 135–143. https://doi.org/10.1016/s0959-6526(98)00051-1.

BFW, 2022. *waldinventur*. Bundesforschungszentrum Für Wald. https://www.waldinventur.at/.

Blank, J., Deb, K., 2020. Pymoo: multi-objective optimization in Python. IEEE Access 8, 89497–89509. https://doi.org/10.1109/ACCESS.2020.2990567.

Blengini, G., Blagoeva, D., Dewulf, J., Torres de Matos, C., Nita, V., Vidal-Legaz, B., Latunussa, C., Kayam, Y., Talens Peirò, L., Baranzelli, C., Manfredi, S., Mancini, L., Nuss, P., Marmier, A., Alves-Dias, P., Pavel, C., Tzimas, E., Mathieux, F., Pennington, D., Ciupagea, C., 2017. Assessment of the Methodology for Establishing the EU List of Critical Raw Materials - Background Report. Publications Office of the European Union. https://doi.org/10.2760/130462.

Boiger, T., Mair-Bauernfeind, C., Asada, R., Stern, T., 2024a. Shifting wood between material and energy use: Modeling the effects of substitution. J. Ind. Ecol. 28, 1198–1211. https://doi.org/10.1111/jiec.13530.

Boiger, T., Mair-Bauernfeind, C., Asada, R., Stern, T., 2024b. Carbon-efficient wood utilisation in times of climate change. Schweizerische Zeitschrift Fur Forstwesen 175 (1), 28–35. https://doi.org/10.3188/szf.2024.0028.

Braun, M., Winner, G., Schwarzbauer, P., Stern, T., 2016. Apparent half-life-dynamics of harvested wood products (HWPs) in Austria: development and analysis of weighted time-series for 2002 to 2011. For. Pol. Econ. 63, 28–34. https://doi.org/10.1016/j.forpol.2015.11.008.

Brunet-Navarro, P., Jochheim, H., Cardellini, G., Richter, K., Muys, B., 2021. Climate mitigation by energy and material substitution of wood products has an expiry date. J. Clean. Prod. 303. https://doi.org/10.1016/j.jclepro.2021.127026.

Brunet-Navarro, P., Jochheim, H., Muys, B., 2016. Modelling carbon stocks and fluxes in the wood product sector: a comparative review. Global Change Biol. 22 (7), 2555–2569. https://doi.org/10.1111/gcb.13235.

Brunet-Navarro, P., Jochheim, H., Muys, B., 2017. The effect of increasing lifespan and recycling rate on carbon storage in wood products from theoretical model to application for the European wood sector. Mitig. Adapt. Strategies Glob. Change 22 (8), 1193–1205. https://doi.org/10.1007/s11027-016-9722-z.

Cambero, C., Sowlati, T., Pavel, M., 2016. Economic and life cycle environmental optimization of forest-based biorefinery supply chains for bioenergy and biofuel production. Chem. Eng. Res. Des. 107, 218–235. https://doi.org/10.1016/j.cherd.2015.10.040.

Cordier, S., Robichaud, F., Blanchet, P., Amor, B., 2021. Regional environmental life cycle consequences of material substitutions: the case of increasing wood structures for non-residential buildings. J. Clean. Prod. 328. https://doi.org/10.1016/j.jclepro.2021.129671.

Dasgupta, P., Chakrabarti, P., Parul, DeSarkar, S.C., 1999. Multiobjective Heuristic Search : an Introduction to Intelligent Search Methods for Multicriteria Optimization. Vieweg & Sohn Verlagsgesellschaft mbH.

Deb, K., Pratap, A., Agarwal, S., Meyarivan, T., 2002. A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Trans. Evol. Comput. 6 (2), 182–197. https://doi.org/10.1109/4235.996017.

Dugan, A.J., Birdsey, R., Mascorro, V.S., Magnan, M., Smyth, C.E., Olguin, M., Kurz, W.A., 2018. A systems approach to assess climate change mitigation options in landscapes of the United States forest sector. Carbon Bal. Manag. 13 (1), 13. https://doi.org/10.1186/s13021-018-0100-x.

European Commission, 2018. A Sustainable Bioeconomy for Europe: Strengthening the Connection between Economy. society and the environment. https://doi.org/10.2777/478385.

European Commission Directorate-General for Research and Innovation, 2012. Innovating for sustainable growth - a bioeconomy for Europe. https://data.europa.eu/doi/10.2777/6462.

European Union, 2020. Long-term low greenhouse gas emission development strategy of the European Union and its Member States. https://unfccc.int/documents/210328.

Federal Ministry Republic of Austria, 2019. Bioeconomy - a strategy for Austria. https://www.bmbwf.gv.at/en/Topics/Research/Research-in-Austria/Strategic-focus-and-advisory-bodies/Strategies/Bioeconomy-Strategy.html.

Fuchs, R., Herold, M., Verburg, P.H., Clevers, J.G.P.W., 2013. A high-resolution and harmonized model approach for reconstructing and analysing historic land changes in Europe. Biogeosciences 10 (3), 1543–1559. https://doi.org/10.5194/bg-10-1543-2013.

Garcia, C.A., Hora, G., 2017. State-of-the-art of waste wood supply chain in Germany and selected European countries. In: Waste Management, vol. 70. Elsevier Ltd, pp. 189–197. https://doi.org/10.1016/j.wasman.2017.09.025.

Geng, A., Yang, H., Chen, J., Hong, Y., 2017. Review of carbon storage function of harvested wood products and the potential of wood substitution in greenhouse gas mitigation. For. Pol. Econ. 85 (159), 192–200. https://doi.org/10.1016/j.forpol.2017.08.007.

Grassi, G., Fiorese, G., Pilli, R., Jonsson, K., Blujdea, V., Korosuo, A., Vizzarri, M., 2021. In: Sanchez Lopez, J., Jasinevičius, G., Avraamides, M. (Eds.), Brief on the Role of the Forest-Based Bioeconomy in Mitigating Climate Change through Carbon Storage and Material Substitution (s), European Commission. JRC124374.

Haberl, H., Geissler, S., 2000. Cascade utilization of biomass: strategies for a more efficient use of a scarce resource. Ecol. Eng. 16, 111–121. https://doi.org/10.1016/s0925-8574(00)00059-8.

Harmon, M.E., 2019. Have product substitution carbon benefits been overestimated? A sensitivity analysis of key assumptions. Environ. Res. Lett. 14 (6), 065008. https://doi.org/10.1088/1748-9326/ab1e95.

Hennigar, C.R., MacLean, D.A., Amos-Binks, L.J., 2008. A novel approach to optimize management strategies for carbon stored in both forests and wood products. For. Ecol. Manag. 256 (4), 786–797. https://doi.org/10.1016/j.foreco.2008.05.037.

Hetemäki, L., Kangas, J., Peltola, H. (Eds.), 2022. Forest Bioeconomy and Climate Change, vol. 42. Springer International Publishing. https://doi.org/10.1007/978-3-030-99206-4.

Hiltunen, M., Strandman, H., Kilpeläinen, A., 2021. Optimizing forest management for climate impact and economic profitability under alternative initial stand age structures. Biomass Bioenergy 147. https://doi.org/10.1016/j.biombioe.2021.106027.

Höglmeier, K., Steubing, B., Weber-Blaschke, G., Richter, K., 2015. LCA-based optimization of wood utilization under special consideration of a cascading use of wood. J. Environ. Manag. 152, 158–170. https://doi.org/10.1016/j.jenvman.2015.01.018.

Homer, J., 2019. Best practices in system dynamics modeling, revisited: a practitioner's view. Syst. Dynam. Rev. 35 (2), 177–181. https://doi.org/10.1002/sdr.1630.

Hurmekoski, E., Myllyviita, T., Seppälä, J., Heinonen, T., Kilpeläinen, A., Pukkala, T., Mattila, T., Hetemäki, L., Asikainen, A., Peltola, H., 2020. Impact of structural changes in wood-using industries on net carbon emissions in Finland. J. Ind. Ecol. 24 (4), 899–912. https://doi.org/10.1111/jiec.12981.

Hurmekoski, E., Smyth, C.E., Stern, T., Verkerk, P.J., Asada, R., 2021. Substitution impacts of wood use at the market level: a systematic review. Environ. Res. Lett. 16 (12). https://doi.org/10.1088/1748-9326/ac386f.

Jasinevičius, G., Lindner, M., Verkerk, P., Aleinikovas, M., 2017. Assessing impacts of wood utilisation scenarios for a Lithuanian bioeconomy: impacts on carbon in forests and harvested wood products and on the socio-economic performance of the forest-based sector. Forests 8 (4), 133. https://doi.org/10.3390/f8040133.

Kalt, G., 2018. Carbon dynamics and GHG implications of increasing wood construction: long-term scenarios for residential buildings in Austria. Carbon Manag. 9 (3), 265–275. https://doi.org/10.1080/17583004.2018.1469948.

Keller, A., 2017. Mathematical Optimization Terminology : a Comprehensive Glossary of Terms. Elsevier/Academic Press.

Keller, A., 2019. Multi-objective optimization in theory and practice. In: Metaheuristic Algorithms, vol. 2. Bentham Science Publishers.

Khmara, Y., Kronenberg, J., 2020. Degrowth in the context of sustainability transitions: in search of a common ground. J. Clean. Prod. 267, 122072. https://doi.org/10.1016/j.jclepro.2020.122072.

Knauf, M., Köhl, M., Mues, V., Olschofsky, K., Frühwald, A., 2015. Modeling the CO2-effects of forest management and wood usage on a regional basis. Carbon Bal. Manag. 10 (1), 13. https://doi.org/10.1186/s13021-015-0024-7.

Kochenderfer, M., Wheeler, T., 2019. Algorithms for Optimization. The MIT Press. The MIT Press.

Koh, L.P., Ghazoul, J., 2008. Biofuels, biodiversity, and people: understanding the conflicts and finding opportunities. Biol. Conserv. 141 (10), 2450–2460. https://doi.org/10.1016/j.biocon.2008.08.005.

Laganovska, I., Feofilovs, M., Blumberga, D., 2022. Forestry sector resource optimization with TIMES. Environmental and Climate Technologies 26 (1), 1279–1298. https://doi.org/10.2478/rtuect-2022-0097.

Lauri, P., Forsell, N., Di Fulvio, F., Snäll, T., Havlik, P., 2021. Material substitution between coniferous, non-coniferous and recycled biomass – impacts on forest industry raw material use and regional competitiveness. For. Pol. Econ. 132, 102588. https://doi.org/10.1016/j.forpol.2021.102588.

Leskinen, P., Cardellini, G., González García, S., Hurmekoski, E., Sathre, R., Seppälä, J., Smyth, C.E., Stern, T., Verkerk, H., 2018. Substitution effects of wood-based products in climate change mitigation. Sci. Pol. 7, 28. https://www.efi.int/publications-bank/substitution-effects-wood-based-products-climate-change-mitigation.

Leturcq, P., 2020. GHG displacement factors of harvested wood products: the myth of substitution. Sci. Rep. 10 (1), 1–9. https://doi.org/10.1038/s41598-020-77527-8.

Liu, Q., Li, X., Liu, H., Guo, Z., 2020. Multi-objective metaheuristics for discrete optimization problems: a review of the state-of-the-art. In: Applied Soft Computing Journal, vol. 93. Elsevier Ltd, 106382. https://doi.org/10.1016/j.asoc.2020.106382.

Liu, X., Bakshi, B.R., 2018. Extracting heuristics for designing sustainable built environments by coupling multiobjective evolutionary optimization and machine learning. In: Computer Aided Chemical Engineering, vol. 44. Elsevier B.V, pp. 2539–2544. https://doi.org/10.1016/B978-0-444-64241-7.50418-3.

Macqueen, J., 1967. Some methods for classification and analysis of multivariate observations. Berkeley Symp. on Math. Statist. and Prob 281–297.

Mair, C., Stern, T., 2017. Cascading utilization of wood: a matter of circular economy?. In: Current Forestry Reports, vol. 3. Springer International Publishing, pp. 281–295. https://doi.org/10.1007/s40725-017-0067-y, 4.

Mehr, J., Vadenbo, C., Steubing, B., Hellweg, S., 2018. Environmentally optimal wood use in Switzerland—investigating the relevance of material cascades. Resour. Conserv. Recycl. 131, 181–191. https://doi.org/10.1016/j.resconrec.2017.12.026.

Myllyviita, T., Hurmekoski, E., Kunttu, J., 2022. Substitution impacts of Nordic wood-based multi-story building types: influence of the decarbonization of the energy sector and increased recycling of construction materials. Carbon Bal. Manag. 17 (1). https://doi.org/10.1186/s13021-022-00205-x.

Myllyviita, T., Soimakallio, S., Judl, J., Seppälä, J., 2021. Wood substitution potential in greenhouse gas emission reduction–review on current state and application of displacement factors. In: Forest Ecosystems, vol. 8. SpringerOpen, p. 42. https://doi.org/10.1186/s40663-021-00326-8, 1.

Nainggolan, R., Perangin-Angin, R., Simarmata, E., Tarigan, A.F., 2019. Improved the performance of the K-means cluster using the sum of squared error (SSE) optimized by using the elbow method. J. Phys. Conf. 1361 (1). https://doi.org/10.1088/1742-6596/1361/1/012015.

Pieragostini, C., Mussati, M.C., Aguirre, P., 2012. On process optimization considering LCA methodology. In: Journal of Environmental Management, vol. 96, pp. 43–54. https://doi.org/10.1016/j.jenvman.2011.10.014, 1.

Poudel, B.C., Sathre, R., Gustavsson, L., Bergh, J., 2011. Climate change mitigation through increased biomass production and substitution: a case study in morth-Central Sweden. In: Proceedings of the World Renewable Energy Congress – Sweden, vol. 57, pp. 628–635. https://doi.org/10.3384/ecp11057628. *8–13 May, 2011, Linköping, Sweden*.

Pukkala, T., 2011. Optimizing forest management in Finland with carbon subsidies and taxes. For. Pol. Econ. 13 (6), 425–434. https://doi.org/10.1016/j.forpol.2011.06.004.

Raymer, A.K., Gobakken, T., Solberg, B., Hoen, H.F., Bergseng, E., 2009. A forest optimisation model including carbon flows: application to a forest in Norway. For. Ecol. Manag. 258 (5), 579–589. https://doi.org/10.1016/j.foreco.2009.04.036.

Rüter, S., Matthews, R.W., Lundblad, M., Sato, A., Hassan, R.A., 2019. Harvested wood products. In: 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, pp. 1–49.

Rüter, S., Werner, F., Forsell, N., Prins, C., Vial, E., Levet, A.-L., 2016. ClimWood2030 - Climate Benefits of Material Substitution by Forest Biomass and Harvested Wood Products: Perspective 2030. Final Report. https://doi.org/10.3220/rep1468328990000.

Santos, A., Carvalho, A., Barbosa-Póvoa, A.P., Marques, A., Amorim, P., 2019. Assessment and optimization of sustainable forest wood supply chains – a systematic literature review. For. Pol. Econ. 105 (February), 112–135. https://doi.org/10.1016/j.forpol.2019.05.026.

Sathre, R., Gustavsson, L., 2009. A state-of-the-art review of energy and climate effects of wood product substitution. Ecotechnology 57, 59.

Sathre, R., O'Connor, J., 2010. Meta-analysis of greenhouse gas displacement factors of wood product substitution. Environ. Sci. Pol. 13 (2), 104–114. https://doi.org/10.1016/j.envsci.2009.12.005.

Schulte, M., Jonsson, R., Eggers, J., Hammar, T., Stendahl, J., Hansson, P.A., 2023. Demand-driven climate change mitigation and trade-offs from wood product substitution: the case of Swedish multi-family housing construction. J. Clean. Prod. 421. https://doi.org/10.1016/j.jclepro.2023.138487.

Seidl, R., Rammer, W., Jäger, D., Currie, W.S., Lexer, M.J., 2007. Assessing trade-offs between carbon sequestration and timber production within a framework of multi-purpose forestry in Austria. For. Ecol. Manag. 248 (1–2), 64–79. https://doi.org/10.1016/j.foreco.2007.02.035.

Seidl, R., Thom, D., Kautz, M., Martin-Benito, D., Peltoniemi, M., Vacchiano, G., Wild, J., Ascoli, D., Petr, M., Honkaniemi, J., Lexer, M.J., Trotsiuk, V., Mairota, P., Svoboda, M., Fabrika, M., Nagel, T.A., Reyer, C.P.O., 2017. Forest disturbances under climate change. Nat. Clim. Change 7 (6), 395–402. https://doi.org/10.1038/nclimate3303.

Shabani, N., Akhtari, S., Sowlati, T., 2013. Value chain optimization of forest biomass for bioenergy production: a review. In: Renewable and Sustainable Energy Reviews, vol. 23. Elsevier Ltd, pp. 299–311. https://doi.org/10.1016/j.rser.2013.03.005.

Shabani, N., Sowlati, T., 2013. A mixed integer non-linear programming model for tactical value chain optimization of a wood biomass power plant. Appl. Energy 104, 353–361. https://doi.org/10.1016/j.apenergy.2012.11.013.

Sharma, S., Kumar, V., 2022. A comprehensive review on multi-objective optimization techniques: past, present and future. Arch. Comput. Methods Eng. 29 (7), 5605–5633. https://doi.org/10.1007/s11831-022-09778-9.

She, J., Chung, W., Han, H., 2019. Economic and environmental optimization of the forest supply chain for timber and bioenergy production from beetle-killed forests in Northern Colorado. Forests 10 (8), 689. https://doi.org/10.3390/f10080689.

Soimakallio, S., Kalliokoski, T., Lehtonen, A., Salminen, O., 2021. On the trade-offs and synergies between forest carbon sequestration and substitution. Mitig. Adapt. Strategies Glob. Change 26 (1), 1–17. https://doi.org/10.1007/s11027-021-09942-9.

Statistik Austria, 2018. Gütereinsatz. https://www.statistik.at/statistiken/industrie-bau-handel-und-dienstleistungen/gueterproduktion/guetereinsatz.

Sterman, J., 2000. Business Dynamics, System Thinking and Modeling for a Complex World. McGraw-Hill higher education. https://www.researchgate.net/publication/44827001_Business_Dynamics_System_Thinking_and_Modeling_for_a_Complex_World .

Strimitzer, L., Höher, M., Nemestothy, K., 2020. Holzströme in Österreich, vols. 1–3.

Suter, F., Steubing, B., Hellweg, S., 2017. Life cycle impacts and benefits of wood along the value chain: the case of Switzerland. J. Ind. Ecol. 21 (4), 874–886. https://doi.org/10.1111/jiec.12486.

Taeroe, A., Mustapha, W.F., Stupak, I., Raulund-Rasmussen, K., 2017. Do forests best mitigate CO2 emissions to the atmosphere by setting them aside for maximization of carbon storage or by management for fossil fuel substitution? J. Environ. Manag. 197, 117–129. https://doi.org/10.1016/j.jenvman.2017.03.051.

Taskhiri, M.S., Jeswani, H., Geldermann, J., Azapagic, A., 2019. Optimising cascaded utilisation of wood resources considering economic and environmental aspects. Comput. Chem. Eng. 124, 302–316. https://doi.org/10.1016/j.compchemeng.2019.01.004.

Wang, R., Haller, P., 2024. Enhancing wood efficiency through comprehensive wood flow analysis: methodology and strategic insights. Forest Ecosystems 11, 100179. https://doi.org/10.1016/j.fecs.2024.100179.

Werner, F., Taverna, R., Hofer, P., Richter, K., 2005. Carbon pool and substitution effects of an increased use of wood in buildings in Switzerland: first estimates. Ann. For. Sci. 62 (8), 889–902. https://doi.org/10.1051/forest:2005080.

Wolf, C., Klein, D., Richter, K., Weber-Blaschke, G., 2016. Mitigating environmental impacts through the energetic use of wood: regional displacement factors generated by means of substituting non-wood heating systems. Sci. Total Environ. 569–570, 395–403. https://doi.org/10.1016/j.scitotenv.2016.06.021.

Zeug, W., Bezama, A., Thrän, D., 2022. Application of holistic and integrated LCSA: case study on laminated veneer lumber production in Central Germany. Int. J. Life Cycle Assess. 27 (12), 1352–1375. https://doi.org/10.1007/s11367-022-02098-x.

Zhang, M., Wang, G., Gao, Y., Wang, Z., Mi, F., 2017. Trade-offs between economic and environmental optimization of the forest biomass generation supply chain in inner Mongolia, China. Sustainability 9 (11), 2030. https://doi.org/10.3390/su9112030.